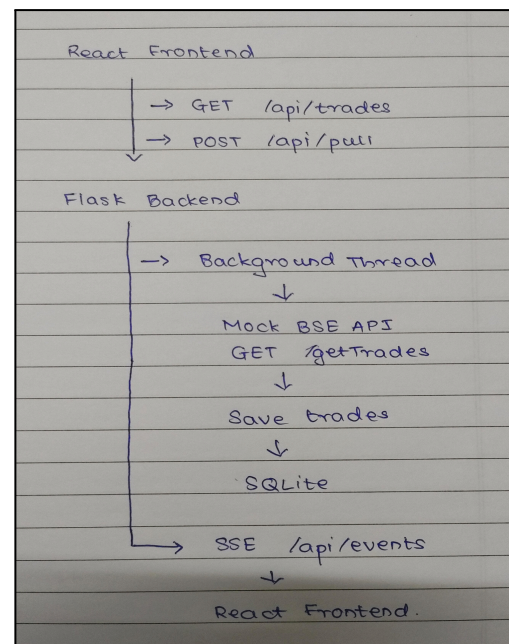


Request Flow

1. React loads existing trades from the Flask backend.
2. User clicks **Pull Latest Trades**.
3. Flask starts the BSE request in a background thread and responds immediately.
4. The background thread calls the delayed Mock BSE API.
5. Received trades are stored in SQLite.
6. Backend sends a pull_completed event through Server-Sent Events (SSE).
7. React receives the event and fetches the latest trades.
8. The dashboard updates without a page refresh or polling.



Design Decisions

Background Thread

A Python thread is used to prevent the long-running BSE request from blocking the Flask API. For this assignment, a thread is simple and sufficient. A production system could use Celery or RQ with Redis for more reliable background job processing.

Server-Sent Events

SSE is used because the backend only needs to notify the browser when the background pull completes. It avoids continuous polling and is simpler than WebSockets for this use case.

SQLite

SQLite provides simple persistent storage for the trades without requiring a separate database server.

Mock BSE API

The Mock BSE service simulates the external BSE API and provides a configurable delay to demonstrate long-running requests.